

Feature-Resolved Attention: A Clean Derivation

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Abstract

Feature-resolved attention (FRA) decomposes an attention circuit in the basis of sparse autoencoder (SAE) features. The goal is to assign a scalar target, such as a downstream feature preactivation or steering direction, to specific query, key, and value-side feature contributions. This note gives a clean derivation of the two pieces we use most often: frozen-attention OV attribution and first-order QK attribution via the softmax Jacobian. It also records the distinction between feature ranking and intervention path, since conflating these two choices has been a recurring source of confusion.

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1 Roadmap

The derivation separates three questions.

1. **What scalar are we explaining?** We choose a target direction $d \in \mathbb{R}^{d_{\text{model}}}$ and explain

$$T_q(d) := \langle \text{attn_out}_q, d \rangle.$$

In the sleeper setting, d can be the encoder direction of a downstream suppressor feature at `hook_resid_mid`.

2. **How do features contribute through a fixed attention pattern?** Holding A fixed makes the OV path linear. This gives a clean per-head, per-source-token, per-feature attribution.
3. **How do features change the attention pattern?** Attention patterns are nonlinear in QK scores. We therefore decompose the scores exactly in feature pairs and then use a first-order softmax-Jacobian approximation to attribute changes in the scalar target.

Main caution. An FRA *ranking* and an FRA *intervention path* are separate design choices. For example, one can rank features by QK attribution but intervene through the V path, the QK path, or the original normalized residual hook.

2 Transformer and Hook Conventions

We use the TransformerLens row-vector convention: activations multiply weights on the right. Thus an activation update has the form

$$\text{new_activation} = \text{old_activation} W + b.$$

Fix a layer ℓ , head h , query position q , and source/key position k . Let

$$x_t \in \mathbb{R}^{d_{\text{model}}}$$

denote the exact tensor read by the attention sublayer at position t . In TransformerLens terms this is typically the layer-normalized attention input, for example `blocks. $\ell$ .ln1.hook_normalized`, after whatever folding convention the model uses.

For head h , define

$$Q_t^h = x_t W_Q^h, \quad K_t^h = x_t W_K^h, \quad V_t^h = x_t W_V^h + b_V^h.$$

The pre-softmax score and attention pattern are

$$s_{qk}^h = \frac{Q_q^h \cdot K_k^h}{\sqrt{d_{\text{head}}}}, \quad A_{qk}^h = \text{softmax}_k(s_{qk}^h).$$

The head output written to the residual stream is

$$y_q^h = \sum_k A_{qk}^h V_k^h W_O^h = \sum_k A_{qk}^h x_k W_V^h W_O^h + c^h, \quad c^h := b_V^h W_O^h.$$

The full attention write is

$$\text{attn_out}_q = \sum_h y_q^h + b_O.$$

If we are targeting a direction at `hook_resid_mid`, then

$$r_{\ell,q}^{\text{mid}} = r_{\ell,q}^{\text{pre}} + \text{attn_out}_{\ell,q},$$

so the downstream feature preactivation decomposes into a skip term plus an attention term:

$$\langle r_{\ell,q}^{\text{mid}}, d \rangle = \langle r_{\ell,q}^{\text{pre}}, d \rangle + \langle \text{attn_out}_{\ell,q}, d \rangle.$$

The FRA derivation below focuses on the attention term.

3 Feature Decomposition

Assume an SAE or dictionary on the attention input x_t :

$$x_t = \sum_{\lambda=1}^{d_{\text{SAE}}} u_t^\lambda f_\lambda + e_t.$$

Here

- u_t^λ is feature λ 's activation at position t ,
- $f_\lambda \in \mathbb{R}^{d_{\text{model}}}$ is the decoder direction,
- e_t is reconstruction error.

We will often drop e_t in the algebra and then treat reconstruction error as an empirical error term.

Remark 1. The same feature basis can be used for Q, K, and V because they read the same attention-input tensor x_t . But the channel-specific vectors

$$f_\lambda W_Q^h, \quad f_\lambda W_K^h, \quad f_\lambda W_V^h W_O^h$$

are different. FRA is useful precisely because it keeps these channels separate.

4 Frozen-Attention OV Attribution

Start with the scalar target

$$T_q(d) := \langle \text{attn_out}_q, d \rangle.$$

For one head, holding A^h fixed gives

$$\langle y_q^h, d \rangle = \sum_k A_{qk}^h \langle x_k W_V^h W_O^h, d \rangle + \langle c^h, d \rangle.$$

Define the virtual OV read direction $u_h^{\text{OV}}(d)$ by

$$\langle x_k W_V^h W_O^h, d \rangle = \langle x_k, u_h^{\text{OV}}(d) \rangle.$$

With row-vector activations, this is equivalently

$$u_h^{\text{OV}}(d) = W_V^h W_O^h d^\top$$

up to the usual row/column identification.

Substitute the feature expansion of x_k :

$$\langle x_k, u_h^{\text{OV}}(d) \rangle \approx \sum_\lambda u_k^\lambda \underbrace{\langle f_\lambda W_V^h W_O^h, d \rangle}_{:= \beta_{h,\lambda}(d)}.$$

Definition 1 (OV feature coefficient). For target direction d , define

$$\beta_{h,\lambda}(d) := \langle f_\lambda W_V^h W_O^h, d \rangle.$$

Proposition 1 (Frozen-attention OV attribution). *Given the observed attention pattern A , the feature-resolved OV contribution of source feature λ at position k , read by query position q through head h , is*

$$C_{h,q,k,\lambda}^{\text{OV}}(d) := A_{qk}^h u_k^\lambda \beta_{h,\lambda}(d).$$

Thus

$$T_q(d) \approx \sum_h \sum_k \sum_\lambda C_{h,q,k,\lambda}^{\text{OV}}(d) + \text{const} + \text{reconstruction error}.$$

Typical feature ranking. For a prompt distribution $\mathcal{D}$, define an OV score

$$\text{Score}_{\text{OV}}(\lambda) = \sum_h \beta_{h,\lambda}(d) \mathbb{E}_{b \sim \mathcal{D}, q \in \text{prompt}(b)} \left[\sum_k A_{b,qk}^h u_{b,k}^\lambda \right].$$

Rank features by $|\text{Score}_{\text{OV}}(\lambda)|$, or by a variant that subtracts the same statistic on a control distribution.

TODO: Specify which distribution is used for the final paper: deployment only, deployment minus clean, or a prompt-position-restricted variant.

5 QK Score Decomposition

OV attribution holds A fixed. To ask how features change the attention pattern, begin at the score level:

$$s_{qk}^h = \frac{(x_q W_Q^h) \cdot (x_k W_K^h)}{\sqrt{d_{\text{head}}}}.$$

Using the feature expansion on both query and key positions,

$$x_q \approx \sum_\mu u_q^\mu f_\mu, \quad x_k \approx \sum_\nu u_k^\nu f_\nu.$$

Then

$$s_{qk}^h \approx \sum_{\mu,\nu} u_q^\mu u_k^\nu \omega_{\mu\nu}^{h,\text{QK}} + \text{bias/error},$$

where

$$\omega_{\mu\nu}^{h,\text{QK}} := \frac{(f_\mu W_Q^h) \cdot (f_\nu W_K^h)}{\sqrt{d_{\text{head}}}}.$$

Definition 2 (QK score atom). The score-level QK atom for query feature μ and key feature ν is

$$S_{qk}^{h,\mu\nu} := u_q^\mu u_k^\nu \omega_{\mu\nu}^{h,\text{QK}}.$$

Remark 2. The score decomposition is clean. The attention-pattern decomposition is not, because $A_{qk}^h = \text{softmax}_k(s_{qk}^h)$ is nonlinear. This is why QK-side target attribution needs either an exact intervention experiment or a linearization.

6 QK Attribution via the Softmax Jacobian

To attribute changes in the scalar target $T_q(d)$ through QK, hold the OV side fixed and ask how a score perturbation changes the attention-weighted OV read.

Define the fixed-OV source scalar

$$g_k^h(d) := \sum_{\lambda} u_k^{\lambda} \beta_{h,\lambda}(d).$$

Then

$$T_q(d) \approx \sum_h \sum_k A_{qk}^h g_k^h(d) + \text{const.}$$

The softmax Jacobian is

$$\frac{\partial A_{qk'}^h}{\partial s_{qj}^h} = A_{qk'}^h \left(\mathbf{1}_{k'=j} - A_{qj}^h \right).$$

Therefore

$$\frac{\partial T_q(d)}{\partial s_{qj}^h} = A_{qj}^h \left(g_j^h(d) - \bar{g}_q^h(d) \right), \quad \bar{g}_q^h(d) := \sum_k A_{qk}^h g_k^h(d).$$

Definition 3 (Centered OV profile). Define

$$\tilde{g}_{qj}^h(d) := A_{qj}^h \left(g_j^h(d) - \bar{g}_q^h(d) \right).$$

6.1 Single query-feature sensitivity

Remove query-side feature μ at query position q :

$$u_q^{\mu} \mapsto 0.$$

The induced score shift at every key j is

$$\delta s_{qj}^h = -u_q^{\mu} \underbrace{\sum_{\nu} u_j^{\nu} \omega_{\mu\nu}^{h,\text{QK}}}_{:= \kappa_j^{h,\mu}}.$$

Thus the first-order target change is

$$\delta T_q^{\mu}(d) \approx -u_q^{\mu} \sum_h \sum_j \kappa_j^{h,\mu} \tilde{g}_{qj}^h(d).$$

6.2 Single key-feature sensitivity

Remove key-side feature ν at source position k :

$$u_k^{\nu} \mapsto 0.$$

Only score s_{qk}^h changes, with

$$\delta s_{qk}^h = -u_k^{\nu} \underbrace{\sum_{\mu} u_q^{\mu} \omega_{\mu\nu}^{h,\text{QK}}}_{:= \eta_q^{h,\nu}}.$$

Therefore

$$\delta T_{q,k}^{\nu}(d) \approx -u_k^{\nu} \sum_h \eta_q^{h,\nu} \tilde{g}_{qk}^h(d).$$

6.3 Pair sensitivity

For the query-key feature pair (μ, ν) , the corresponding first-order score contribution is

$$u_q^\mu u_k^\nu \omega_{\mu\nu}^{h, \text{QK}}.$$

Removing that atom gives

$$\delta T_{q,k}^{\mu,\nu}(d) \approx -u_q^\mu u_k^\nu \sum_h \omega_{\mu\nu}^{h, \text{QK}} \tilde{g}_{qk}^h(d).$$

Typical QK feature ranking. For query-feature ranking, compute

$$\text{Pred}_{\text{QK}}[\mu](b, q) := u_{b,q}^\mu \sum_h \sum_j \kappa_{b,j}^{h,\mu} \tilde{g}_{b,qj}^h(d),$$

and aggregate over prompt positions:

$$\text{Score}_{\text{QK}}(\mu) = \mathbb{E}_{(b,q) \in \text{prompt}} [|\text{Pred}_{\text{QK}}[\mu](b, q)|].$$

This is the “QK L1 mean” ranking used in the sleeper QK/QK note.

7 Intervention Paths

Attribution gives a ranked feature set F . Intervention path is a separate choice.

Given selected features F , define the SAE feature-removal delta at the attention input:

$$\Delta x_t = \sum_{\lambda \in F} [\text{decode}(z_t \text{ with } z_t^\lambda = 0) - \text{decode}(z_t)] \approx - \sum_{\lambda \in F} u_t^\lambda f_\lambda.$$

Protocol 1 (All-path intervention). Add the delta directly at the attention input:

$$x_t \leftarrow x_t + \alpha \Delta x_t.$$

This changes Q, K, and V simultaneously.

Protocol 2 (OV-only intervention). Project the delta through W_V^h and add it at the value hook:

$$V_t^h \leftarrow V_t^h + \alpha \Delta x_t W_V^h.$$

This changes value vectors but leaves the attention pattern fixed.

Protocol 3 (QK-only intervention). Project the delta through W_Q^h and W_K^h , then add it at the Q and K hooks:

$$Q_t^h \leftarrow Q_t^h + \alpha \Delta x_t W_Q^h, \quad K_t^h \leftarrow K_t^h + \alpha \Delta x_t W_K^h.$$

This changes the attention pattern while leaving values unchanged.

Remark 3. This gives a useful grid: QK-ranked features can be sent through QK, OV, or all paths; OV-ranked features can likewise be sent through QK, OV, or all paths. The labels “QK/QK”, “QK/OV”, and “OV/OV” should therefore be read as “ranking/intervention-path”, not as a single inseparable method.

Math object	Typical tensor/code object	Shape sketch
x_t	<code>ln1.hook_normalized</code>	batch, seq, d_{model}
u_t^λ	SAE encodings z	batch, seq, d_{SAE}
f_λ	<code>sae.W_dec[lambda]</code>	$d_{\text{SAE}}, d_{\text{model}}$
A_{qk}^h	<code>attn_pattern</code>	batch, head, query, key
W_Q^h, W_K^h, W_V^h	<code>model.W_Q[ell]</code> , etc.	head, $d_{\text{model}}, d_{\text{head}}$
Δx_t	<code>compute_sae_delta(...)</code>	batch, seq, d_{model}

Table 1: Mapping between the derivation and typical TransformerLens/SAE tensors.

8 Implementation Correspondence

Minimal QK/QK pseudocode.

```

features = topk(qk_l1_mean_scores, k=K)
delta = sum(compute_sae_delta(feature=f) for f in features)

q_delta = einsum("btd,hdk->bthk", delta, W_Q[layer])
k_delta = einsum("btd,hdk->bthk", delta, W_K[layer])

hook_q: q[:, :P, :, :] += alpha * q_delta
hook_k: k[:, :P, :, :] += alpha * k_delta

```

9 Open Choices for the Paper

1. **Target distribution.** Decide whether the headline ranking uses deployment prompts only, deployment-minus-clean, or a prompt-position mask.
2. **Scalar target.** Decide whether to present d as a downstream SAE encoder direction, a logit direction, or a generic steering direction.
3. **Error terms.** Decide how much of the SAE reconstruction error and LN linearization error should appear in the main text versus an appendix.
4. **Intervention terminology.** Use consistent names:

ranking source/intervention path

for example QK/QK, QK/OV, OV/OV, and QK/All.

A Appendix Placeholders

A.1 LayerNorm linearization

TODO: Insert the resid-pre to ln1 two-stage derivation if this writeup needs the sleeper-specific two-stage path.

A.2 Channel-wise interaction decomposition

TODO: Insert the Q/K/V coalition or Harsanyi decomposition if the paper needs singles, pairs, and triples rather than the frozen-OV/QK split.

A.3 Experimental protocol

TODO: Insert exact prompts, feature counts, alpha sweeps, and metrics for each model family.